

Hariprasad Ravishankar

PRINCIPAL ENGINEER & SENIOR TEAM LEAD AT MATHWORKS

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WORK EXPERIENCE

MathWorks, Principal Engineer & Senior Team Lead

May 2026 - Present

- Guide a high-performing compiler team towards advanced AI compilation solutions, focusing on MLIR/CGIR pipeline optimizations and SIMD/CUDA kernel efficiency.
- Launched the **MATLAB Coder Support Package for PyTorch and LiteRT Models**, enabling direct, production-grade C++/CUDA code generation from PyTorch and TFLite models.
- Maintainer of **Predict**, **Stateful Predict**, **Stateful Classify**, and **Classify** blocks in the Deep Learning Toolbox's Simulink library, supporting robust deployment pipelines.
- Optimize deep learning compiler code generation and execution utilizing CUDA, cuBLAS, cuDNN, and TensorRT to maximize hardware utilization.

MathWorks, Senior Team Lead & Senior Software Engineer

2019 - May 2026

- Led a team of 3 high-performing software engineers managing technical execution, mentoring, and professional development.
- Defined and drove the product roadmap for deep learning compiler technologies, presenting strategy and progress to **Vice Presidents** and executive leadership bi-annually.
- Architected end-to-end C++/CUDA code generation from PyTorch, TFLite, and MATLAB frameworks.
- Drove the technical integration of deep learning code generation into **Simulink**, supporting the **Predict**, **LiteRT**, and **PyTorch** blocks with full compatibility for **Rapid Acceleration**, **Model Reference Acceleration**, and advanced cache handling.
- Optimized compiler-generated code for deep learning models using CUDA, cuBLAS, cuDNN, and TensorRT libraries.

MathWorks, Software Engineer in Test

2018 - 2019

- Responsible for enhancing testability, developing test suites, and conducting hands-on testing for GPU code-generation and deep learning deployment capabilities.
- Tested deployment of Deep Learning networks to Embedded targets like GPUs, FPGAs, and ARM processors.
- Tested code-generation engine from MATLAB & Simulink, code-generation optimizations, and interfacing generated code with hardware targets.

MathWorks, Application Support Engineer

2017 - 2018

- Technical specialist in MATLAB and MATLAB products in Neural Networks, Image Processing, Deep Learning, Machine Learning, and Optimization algorithms.
- Solved complex technical issues for users in academia and industry, collaborating with development.
- Supported development in identifying quality issues, testing performance enhancements, and improving customer deep learning workflows.

NinePoint Medical, Deep Learning Co-op

2017

- Developed Deep Learning Architecture for efficient segmentation of clinical features in OCT images using TensorFlow.
- Designed custom networks based on well-known architectures such as U-Net and Fully Convolutional Net for image segmentation.
- Developed verification scripts and post-processing algorithms for offline deployment.
- Feature pushed to production with official **FDA clearance**.

Philips Research, Summer Research Intern

Jun 2016 - Jul 2016

- Worked on a Deep Learning Approach for Nucleus Segmentation in Cervical Cancer Images.
- Implemented **Segmentation by Classification** and **Segmentation using Deconvolution** techniques using DIGITS by NVIDIA.
- Designed and trained an architecture based on AlexNet and a 2-stage Deconvolution network for segmentation.
- Detected and segmented unseen nuclei with a PPV of 91% and Jaccard Score of 95%, making it viable for commercial deployment.

EDUCATION

MS in Electrical Engineering

, University of Southern California

GPA
3.86/4.0

B.Tech in Electronics Engineering

, Vellore Institute of Technology

CGPA
9.13/10

SKILLS SUMMARY

Expertise: C++, MATLAB, Simulink, CUDA, Python, PyTorch, LiteRT/TFLite, MLIR, CGIR

Libraries/APIs: TensorRT, cuDNN, cuBLAS, OpenCV

Leadership: Team Management, Roadmap Planning, Executive Presentations